16 (a) Show that the equation $5x = x^3 - 9$ can be rearranged to give $x = \sqrt[3]{9 + 5x}$

$5x = x^3 - 9 \rightarrow x^3 = 9 + 5x$

Take cube roots: $x = \sqrt[3]{(9 + 5x)}$

1 mark
Undo the cube.

(1)

(b) Starting with $x_0 = 3$,
use the iteration formula $x_{n+1} = \sqrt[3]{9 + 5x_n}$ three times to find an estimate for a solution of $5x = x^3 - 9$
Give your answer correct to 3 decimal places.

$x1 = \sqrt[3]{(9 + 5 \times 3)}$

1 mark
Substitute x0 = 3.

$= 2.884499141...$

$x2 = \sqrt[3]{(9 + 5 \times 2.884499141...)}$

1 mark
Use the previous result.

$= 2.861174787...$

$x3 = \sqrt[3]{(9 + 5 \times 2.861174787...)}$

$= 2.856418240...$

After 3 iterations: $x \approx 2.856$ (3dp)

1 mark
Round only the final value.

(3)

(Total for Question 16 is 4 marks)

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